

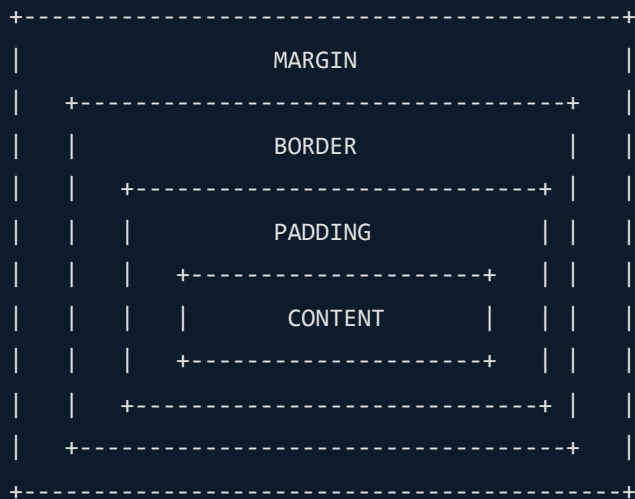
Topic 5: CSS Box Model and Layout

Learning Objectives

- Understand the four layers of the CSS box model
- Use margin, padding, and border correctly
- Apply `box-sizing: border-box` for predictable sizing
- Understand the difference between block and inline elements

5.1 The CSS Box Model

Every HTML element on a page is a rectangular box. The box model describes four layers:



- **Content** — The actual text, image, or other content inside the element
- **Padding** — Transparent space between the content and the border (inside the element)
- **Border** — A visible or invisible line around the padding and content
- **Margin** — Transparent space outside the border (between this element and others)

5.2 Setting Margin, Padding, and Border

```
div {  
  /* Single value – all four sides */  
  padding: 20px;  
  margin: 20px;  
}
```

```

/* Two values – vertical (top/bottom) | horizontal (left/right) */
padding: 10px 20px;
margin: 10px auto; /* auto on sides = center horizontally */

/* Four values – top | right | bottom | left (clockwise) */
padding: 10px 20px 15px 20px;
margin: 5px 10px 5px 10px;

/* Individual sides */
padding-top: 10px;
padding-right: 20px;
padding-bottom: 10px;
padding-left: 20px;

/* Border */
border: 2px solid #333;
border-radius: 8px; /* rounded corners */
}

```

5.3 Width and Height

```

div {
  width: 300px;
  height: 200px;
  max-width: 800px; /* never wider than this */
  min-width: 200px; /* never narrower than this */
}

```

5.4 The box-sizing Problem

By default, `width` only applies to the **content** area. Padding and border are added on top — making the element larger than expected:

```

/* Default: content-box */
div {
  width: 200px;
  padding: 20px;
  border: 5px solid black;

  /* Actual rendered width = 200 + 20+20 + 5+5 = 250px */
}

```

Fix with `border-box` :

```
div {
  width: 200px;
  padding: 20px;
  border: 5px solid black;
  box-sizing: border-box;
  /* Actual rendered width = exactly 200px */
}
```

Best practice — apply globally to every element:

```
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}
```

5.5 The display Property

The `display` property controls how an element behaves in the page flow:

Block elements (`display: block`):

- Start on a new line
- Take the full available width
- You can set width, height, margin, and padding freely
- Default for: `<div>` , `<p>` , `<h1>` – `<h6>` , `<section>` , `<ul>` , `<li>`

Inline elements (`display: inline`):

- Flow with the surrounding text — no new line
- Width and height have no effect
- Horizontal margin and padding work; vertical does not
- Default for: `<span>` , `<a>` , `<strong>` , `<em>` , `<img>`

Inline-block (`display: inline-block`):

- Flows inline (no new line before or after)
- But width, height, margin, and padding all work fully
- Useful for navigation buttons, icon + text combinations

```
/* Side-by-side buttons without flexbox */
.btn {
  display: inline-block;
  width: 120px;
```

```
padding: 10px 16px;
background-color: #3a86ff;
color: white;
text-align: center;
border-radius: 4px;
}
```

5.6 Practical Example

```
<style>
  * { box-sizing: border-box; margin: 0; padding: 0; }

  .card {
    width: 300px;
    padding: 20px;
    margin: 16px auto;
    border: 1px solid #ddd;
    border-radius: 8px;
    background-color: #fff;
  }

  .card h2 {
    font-size: 20px;
    margin-bottom: 8px;
    color: #1d3557;
  }

  .card p {
    color: #555;
    line-height: 1.5;
  }
</style>

<div class="card">
  <h2>Card Title</h2>
  <p>This is a simple card component using the box model.</p>
</div>
```

5.7 Inspecting the Box Model with DevTools

Open browser Developer Tools with **F12**, click on any element, then look at the **Computed** tab. You will see a color-coded box model diagram showing the exact margin, border, padding, and content dimensions.

Key Takeaways

- Every element is a box: content + padding + border + margin
- **Padding** is space inside the element; **Margin** is space outside
- Always apply `box-sizing: border-box` globally with `*` selector
- **Block** elements take full width; **inline** elements flow with text
- Use `display: inline-block` when you need inline flow with width/height control
- Use browser DevTools to inspect and debug box model issues

Reference Materials

- YouTube: CSS Box Model Explained — https://www.youtube.com/watch?v=a_iQb1InAEQ (3:00:00 mark)
- MDN The Box Model — https://developer.mozilla.org/en-US/docs/Learn/CSS/Building_blocks/The_box_model
- W3Schools CSS Box Model — https://www.w3schools.com/css/css_boxmodel.asp

Quiz

Question 1: In the CSS box model, which layer creates space between the element's content and its border?

- A) margin
- B) padding
- C) border
- D) outline

Correct Answer: B — Padding is the space inside the element between the content and the border. Margin is the space outside the border, between this element and neighboring elements.

Question 2: What does applying `box-sizing: border-box` to an element do?

- A) Adds a decorative box-style border
- B) Makes the declared width and height include padding and border
- C) Removes all margins from the element
- D) Turns the element into a flex container

Correct Answer: B — With `border-box`, the `width` and `height` values include padding and border, so the element stays exactly the size you specify without any surprise extra size.

